

EXP.NO: 3

AWKScript Programming

DATE: 30/01/2026

AIM:

To understand and implement AWK scripting for pattern scanning and processing, including field manipulation, arithmetic operations, and conditional logic on text files.

ALGORITHM:

1. Start the experiment.
2. Open the Linux terminal.
3. Create a sample data file using echo or a text editor.
4. Use AWK to print specific fields from the file using field separators.
5. Apply AWK conditions to filter records that match a pattern.
6. Perform arithmetic operations such as sum and average using AWK.
7. Use AWK built-in variables NR (record number) and NF (field count).
8. Use printf in AWK for formatted output.
9. Write an AWK script file (.awk) and execute it with `awk -f`.
10. Stop the experiment.

CODE:

```
# Create sample data file
cat > students.txt << EOF
Alice      85 90 78
Bob        70 65 80
Charlie    92 88 95
Diana      60 75 70
EOF

# Print all lines
awk '{ print }' students.txt

# Print specific fields: name and first mark
awk '{ print $1, $2 }' students.txt

# Print with line numbers using NR
awk '{ print NR, $0 }' students.txt

# Filter students scoring above 80 in first subject
awk '$2 > 80 { print $1, "passed with", $2 }' students.txt

# Calculate total marks for each student
awk '{ total = $2 + $3 + $4
      print $1, "Total:", total }' students.txt

# Calculate average and classify
awk '{
```

```

    avg = ($2 + $3 + $4) / 3
    if (avg >= 80)
        grade = "Distinction"
    else if (avg >= 60)
        grade = "Pass"
    else
        grade = "Fail"
    printf "%-10s Avg: %.1f Grade: %s\n", $1, avg, grade
}' students.txt

# Count total number of records
awk 'END { print "Total students:", NR }' students.txt

```

OUTPUT:

```

[hp@localhost ~]$ awk '{ print $1, $2 }' students.txt
Alice 85
Bob 70
Charlie 92
Diana 60

[hp@localhost ~]$ awk '$2 > 80 { print $1, "passed with", $2 }' students.txt
Alice passed with 85
Charlie passed with 92

[hp@localhost ~]$ awk '{ total=$2+$3+$4; print $1,"Total:",total }' students.txt
Alice Total: 253
Bob Total: 215
Charlie Total: 275
Diana Total: 205

[hp@localhost ~]$ awk '{avg=($2+$3+$4)/3; if(avg>=80) grade="Distinction"; else grade="Pass"; printf "%-10s Avg: %.1f Grade: %s\n", $1, avg, grade}' students.txt
Alice      Avg:84.3 Grade:Distinction
Bob        Avg:71.7 Grade:Pass
Charlie    Avg:91.7 Grade:Distinction
Diana      Avg:68.3 Grade:Pass

[hp@localhost ~]$ awk 'END{print "Total students:", NR}' students.txt
Total students: 4

```

RESULT:

Thus, AWK scripting was successfully used to perform field extraction, conditional arithmetic operations, and formatted output on text data in Linux. filtering,